OS Lab 0 — Image Sharpening: Timings and Analysis

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Assignment summary

We implemented the three stage sharpening pipeline (S1: smoothen, S2: find details, S3: sharpen) and measured durations for five phases: file read (t0), S1 (t1), S2 (t2), S3 (t3), file write (t4). Each experiment was repeated 5 times and the averages (over runs) are presented below. The x axis of the plot is images arranged by their pixel count (width × height).

Image dimensions and areas

Image	Dimensions (W × H)	Area (pixels)
1.ppm	450 × 180	81 000
2.ppm	786 × 393	308 898
3.ppm	1 024 × 575	588 800
4.ppm	1 280 × 910	1 164 800
5.ppm	1 500 × 1 000	1 500 000
6.ppm	2 271 × 1 500	3 406 500
7.ppm	3 750 × 2 470	9 262 500

Averaged timings (seconds)

Image	t0 (read)	t1 (S1)	t2 (S2)	t3 (S3)	t4 (write)	total
1.ppm	0.001401	0.012567	0.003127	0.000000	0.012444	0.029539
2.ppm	0.033959	0.024981	0.015629	0.006425	0.031322	0.112316
3.ppm	0.063209	0.043747	0.028108	0.025326	0.053355	0.213746
4.ppm	0.121892	0.090932	0.056458	0.050298	0.100144	0.419724
5.ppm	0.162890	0.122210	0.062532	0.059383	0.154013	0.561027
6.ppm	0.373209	0.269646	0.134848	0.135230	0.288017	1.200951
7.ppm	1.023692	0.736183	0.372613	0.388722	0.799408	3.320619

Processing (S1+S2+S3) averages: Image1..7 = 0.015694, 0.047035, 0.097181, 0.197688, 0.244125, 0.539724, 1.497518

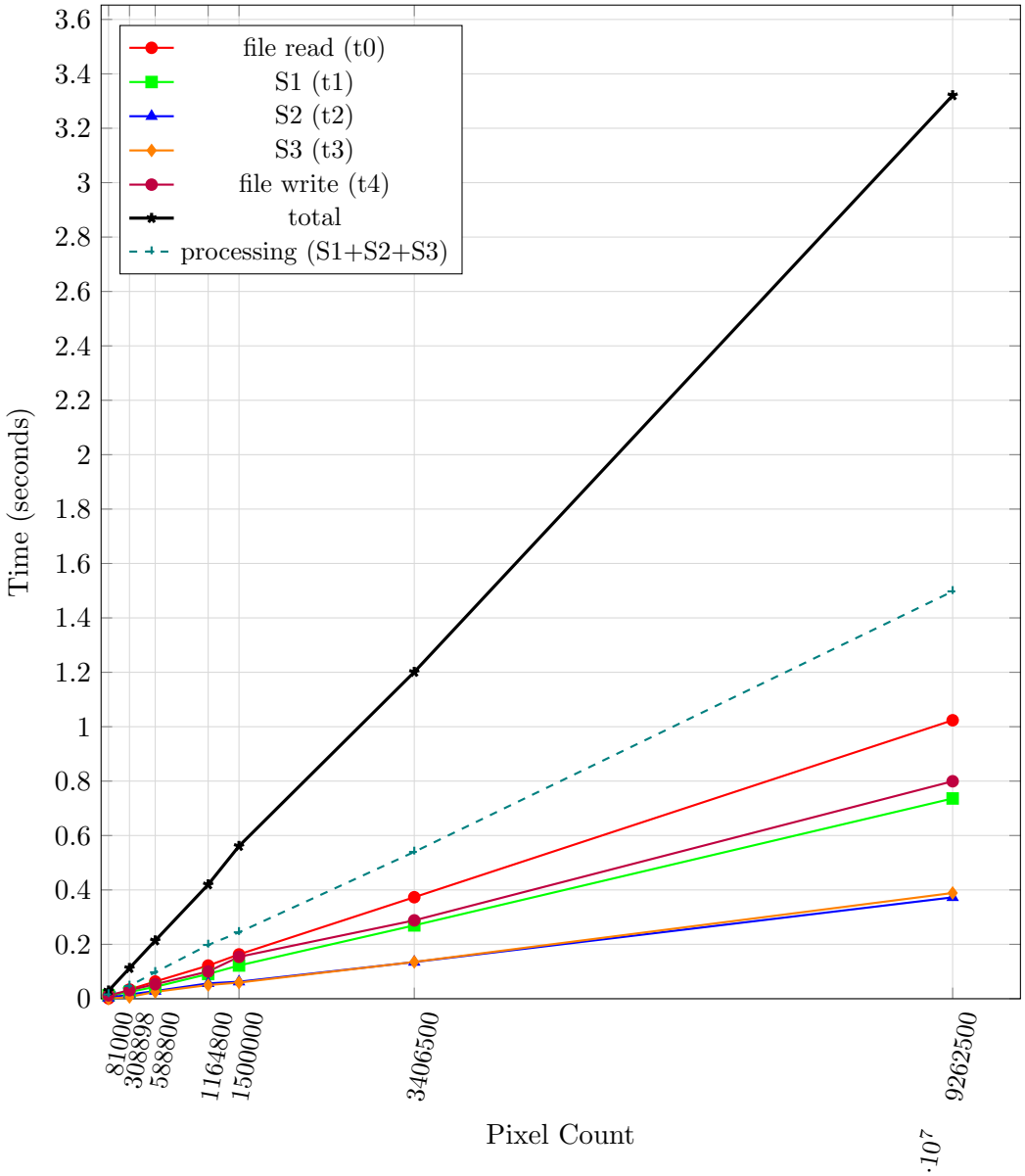


Figure 1: Timings vs image area. Each line is the average over 5 runs.

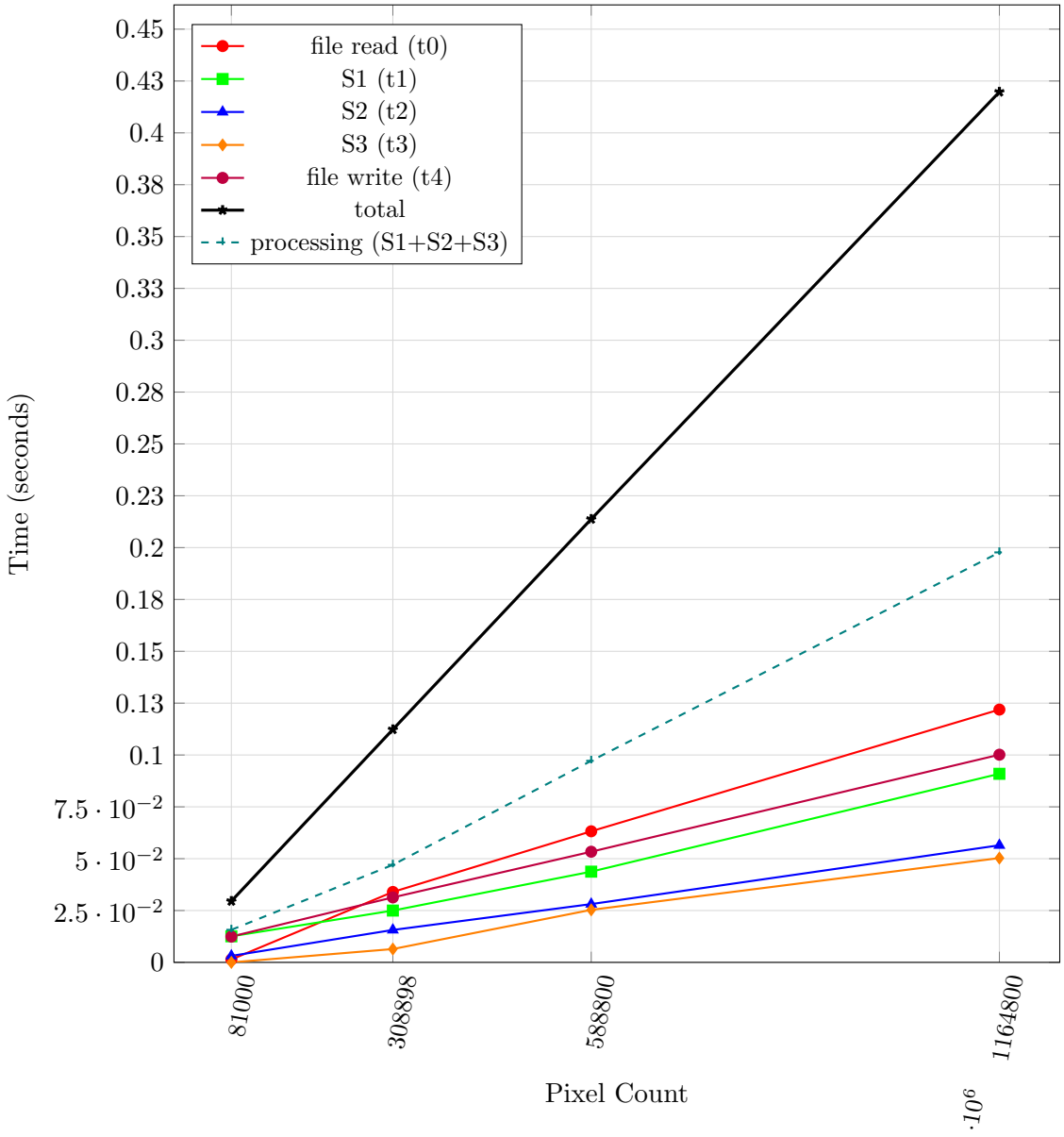


Figure 2: Timings vs image area, Zoomed in to show data for first four images. Each line is the average over 5 runs.

Notes and observations

- Read and write take very less time for small images but significant time for the largest image.
- The processing phase takes lesser time compared to read and write.
- The plotted "processing" line (S1+S2+S3) clearly shows that the three image transformations scale roughly linearly with image size (due to the increase in size of the inages).

Submitted:

A zip file named `CS23BT041_CS23BT028_lab0.zip` containing:

- This report (`CS41_CS28_lab0_report.pdf`)
- The `src` folder containing:
  - The final code files
  - measurement script `getTimesOUT.sh` to run the executable five times for images `1.ppm` to `7.ppm`
  - `makefile` for compiling and running the sharpener and running the time measurement script